Decision log: the embodied-fly path and its branch points

*Every crossroads on the way from the chip-to-animal study to the embodied fly, with the options that were open, the one taken, why, and where to branch from to explore another. Results are not restated here: each stage's numbers live in `report/ROADMAP_EMBODIED_FLY.md` §7, cited to `outputs/metrics.json`, and each stage's registration in `preregistration/`. Commits are on master unless marked otherwise. Decisions taken by the user are marked **(user)**; the rest were taken by the agent and are open to revision.*

How to explore another branch

Check out the commit named in the decision, create a branch, register the alternative in `preregistration/` before running it (fresh seeds if the original result has been seen), and run it with its own command or `python -m neuromps.fly.pipeline --only <step>`.

Decisions

D1. Which fly connectome release the model runs on

- **Options:** FlyWire 630 (the published model's release), FlyWire 783 (the current one).
- **Taken:** 783 for the studies, with a registered 630-vs-783 replication of the published sugar result. **Why:** the current release, with the published release kept as a check.
- **Revisit at:** `fly_chip_to_animal` registration; `neuromps.fly.connectome.load_630`.

D2. Which body simulator for the embodied fly

- **Options:** FlyGym / NeuroMechFly v2 (needs Python 3.12), the FlyMimic musculoskeletal leg loaded in plain MuJoCo, flybody.
- **Taken:** FlyMimic in plain MuJoCo (pinned `mujoco` as the optional `embodied` extra); flybody later for the proboscis. **Why:** muscle-driven, runs in the project's Python 3.11, no new framework. **Revisit:** a FlyGym environment would give six muscle-driven legs only if its musculoskeletal model covers them.

D3. After stage 1 failed: what to change first

- **Options:** rescale the male-CNS synaptic weight further; drive the male nerve cord from descending neurons with the brain removed; switch connectome.
- **Taken:** the nerve cord from descending neurons (stage 1c), then the FlyWire brain joined to it through type-matched descending neurons (stages 1e, 1f). **Why:** the male brain avalanches under the published parameters (`fly_spread`); each connectome is used where its model is validated.

D4. Stage 1e's precondition failed on an absolute rate line

- **Taken:** stage 1f, a relative precondition and depth of reach, on fresh seeds; 1e stays uninterpretable. **Why:** the absolute line had no basis.

D5. Stage 2a failed: how to look for a walking rhythm (user: sweep)

- **Options offered:** fix adaptation parameters from literature; sweep them and report the whole sweep. **Taken (user):** the sweep (stage 2c), with a registered selection rule and confirmation.
- **Revisit:** published adaptation values, if any appear for fly leg or premotor neurons.

D6. The glutamate sign rule (user: go ahead)

- **Taken by the agent:** two bounding rules (glutamate excitatory onto motor neurons only; and everywhere), each with the adaptation grid (stage 2d). **Unexplored:** under the corrected rhythm measure the selection rule would pick (glutamate everywhere, 16 mV, 100 ms); it was not confirmed on fresh seeds.

D7. A different neuron model (user: look at how others make flies walk)

- **Found:** the walking flies online use trained controllers (Eon Systems, flybody); only Pugliese et al. (2025) get a rhythm from nerve-cord wiring, with a firing-rate model.
- **Taken:** reimplement their model from the paper and configuration (their code is unlicensed).
- **Size stand-in (stage 2e):** total synapse count, because the male CNS has no volumes. This starved DNg100; stage 2f drove every command neuron to the same rate; stage 2g used their front-leg network and recruitment rule; stage 2h a geometric input search.
- **Unexplored:** neuron volumes (a future male-CNS release, or MANC volumes through the male-CNS-to-MANC matching); their own network files under a licence.

D8. A flaw in the registered rhythm measure

- **Taken:** a registered audit with a peak-requiring measure (prominence 0.1, calibrated on synthetic signals only), re-scoring stages 1d-2e on their original seeds; registered outcomes kept as recorded, the documents carry both. Stage 1d's D2 withdrawn.
- **Revisit:** prominence 0.05 or 0.2 give the same calibration results; a different peak criterion could be registered.

D9. Where to close the loop first (stage 3)

- **Options:** the front leg (muscle-driven, thinly traced proprioceptors), a middle or hind leg (well traced, but only position-servoed in flybody), the proboscis.
- **Taken:** the front leg through the claw neurons; the unknown flexion/extension tuning bracketed by two assignments and a sham; then a perturbation test (stage 3b).
- **Unexplored:** a hind-leg loop through flybody's servos and its 22 claw neurons; hook and club neurons; campaniform sensilla.

D10. Drugs in the loop (stage 4)

- **Taken:** each knob acts on every network of a trial; movement disruption uses

log2((x + 0.01 mm)/(x_vehicle + 0.01 mm))	with the parent's 0.5 line, 0.05 mm as sensitivity.

- **Disclosed issue:** at full GABA-A block the reach depth is negative, so that condition's index averages the two proboscis terms. **Revisit:** a registered rule for negative depths.

D11. Checking against real flies (stage 5)

- **Taken:** a search protocol registered before any search; every candidate classified.
- **Unexplored:** neuron-specific simulations of the identified-neuron studies (DSOG1, 13A/13B) instead of class-wide knobs; a satiety state for the feeding model.

D12. Release (stage 6)

- **Taken:** a pipeline that reruns every later fly study as its own process, wired into `run_all.py` as a full-run-only `embodied` stage. **Why:** one command reproduces every number; separate processes keep the connectomes from accumulating in memory.

D13. Working overnight without the user (user: continue, commit, push, update docs)

- **Taken:** "push" read as merging finished work into master through a pull request, because GitHub keeps master only; every crossroads tonight is added here.

D14. The precision a chip needs, at the movement level (overnight)

- **Options:** leave stage 4a's timing result as it stands; compute, as the parent study did at the neural level, the chip's change at the dose where movement first fails.
- **Taken:** the latter, as an exploratory analysis (both inputs had been seen), added to the pipeline as `stage4a_precision`. **Why:** the paper's assay-design message rests on the precision a chip needs at behavioural onset; this extends it to movement at no simulation cost.

D15. Confirming the cell the corrected measure selects (overnight)

- **Options:** leave stage 2d's alternative selection unconfirmed (as the audit registered); confirm it on fresh seeds.
- **Taken:** confirm it (registration `embodied_stage2d_confirm_corrected.json`, commit 8495b57), with the corrected measure throughout. **Why:** it is the one loose end the audit named, and it is cheap. **Not taken tonight:** a hind-leg closed loop through flybody's servos (D9), a volume-based size rule (D7), neuron-specific drug simulations (D11), each larger.

D16. Bringing the embodied fly into the technical report and a paper draft (overnight)

- **Taken:** Act VIII of `report/TECHNICAL_REPORT.md` (§27-§31, every number cited, the uncited-number budget unchanged) and `report/PAPER_DRAFT.md` (abstract to methods, every number cited). **Why:** the user asked for a paper and for the report to keep being retouched; the draft is a starting point for the team, not a submission.

D17. Testing the model against identified-neuron experiments (overnight)

- **Options:** a hind-leg closed loop (D9), a volume size rule (D7), simulating the identified neurons that stage 5's published studies manipulated (D11).
- **Taken:** the identified neurons (stage 5b, registration commit c20fa08): the left front leg's GABAergic 13A and 13B hemilineage neurons, silenced (outputs removed) or activated (150 Hz), in the grooming relay of stage 1f. **Why:** it turns stage 5's indirect comparison into a direct one with data already in hand, and it is affordable overnight. **Not possible:** Pool et al.'s DSOG1 neurons are not annotated in FlyWire. **Detail:** 73 of the 76 annotated 13B neurons of that neuromere are in the nerve-cord network used.
- **Revisit:** subsets of 13A/13B (the study manipulated sub-populations), pulsed activation (the study's 70 ms pulses induced grooming-like rhythms), the femur-tibia angle as a readout ("legs locked in flexion").

D18. Reading the 13A/13B manipulations at the joint (overnight)

- **Options:** pulsed 13A activation (the study's grooming-inducing protocol; a leaky integrate-and-fire network has no rebound, so it would very likely fail by construction); a hind-leg closed loop (larger); the joint readout the study actually used.
- **Taken:** the joint readout (stage 5c, registration commit 04a2387, fresh seeds): femur-tibia angle under the same arms as stage 5b. **Why:** stage 5b's mismatch could be a readout mismatch (reach versus joint posture); this tests that directly and cheaply.
- **Process note:** a test edit for this step did not apply and a piped test run hid it; fixed in commit fd51761. Test runs before commits are no longer piped.

D19. Closing the loop on a well-traced leg (overnight)

- **Options:** stop stage 3 at the front-leg limit; close the loop on the right hind leg (25 claw neurons) through flybody's servoed tibia; build a muscle-driven hind leg (no such model at hand).
- **Taken:** the servoed hind leg (stage 3c, registration commit 22b082e): the walking command holds the right hind tibia extended (its extensor pool is tonically active, stage 2a), a 20 ms torque pushes it, and the return is timed under the resistance and assistance assignments and a sham, as in stage 3b. **Why:** it tests whether the front leg's failure was the thin sensor tracing, with the one body that has the leg. **Cost:** feedback changes a commanded angle, not muscle force; a servo's own stiffness sets much of the return.
- **Revisit:** a muscle-driven hind leg (if FlyMimic or another model adds one); the walking command's other legs.

D20. Testing stage 3c's exploratory difference on its own (overnight)

- **Options:** report stage 3c's resistance-against-sham difference as exploratory and stop; test it alone on fresh seeds.
- **Taken:** test it (stage 3d, registration commit 8ea194a): resistance against sham, paired by seed, 32 fresh seeds, a one-sided sign test and a 5 ms margin. **Why:** the difference was the one sign that the push-following part of the feedback acts on a well-traced leg, and it was seen in a noisy, unstable regime, so only a registered test on new seeds can say whether it is real.
- **Outcome:** 0 of 2 (stage 3 stays a documented negative). **Revisit:** a muscle-driven hind leg.

D21. After stage 3: replicate the headline before anything new (overnight)

- **Options:** new science that needs a model not at hand (a muscle-driven hind leg, neuron volumes for the size rule); 13A/13B subsets or pulsed activation (a leaky integrate-and-fire network has no rebound); paper and slide writing only; a registered replication of stage 4a, the paper's headline, which ran on one set of 8 seeds.
- **Taken:** the replication (stage 4b, registration commit b23f6a3): stage 4a's movements on 16 fresh seeds, against the chip detections of the parent design's own fresh-seed replication, with each movement left out in turn. **Why:** it is the result a reviewer will press first, it is cheap (about 15 minutes), and the leave-one-out check names the assumption most likely to make it wrong (that one movement carries the disruption index).
- **Outcome:** 3 of 4 (registration commit b23f6a3). The chip is late in every family on the fresh seeds; without the bitter-gated proboscis GABA-A potentiation ties the chip.
- **Revisit:** a different impairment line for movement (the 0.05 mm sensitivity is reported); a finer dose grid for GABA-A potentiation and excitability block, where the margin is one step.

D22. Resolving stage 4b's one-step margins (overnight)

- **Options:** leave the two thin margins reported; a finer dose grid inside them, as the motor-level tie-break did; a different movement impairment line (a new rule chosen after the results, so not taken).
- **Taken:** the finer grid (stage 4c, registration commit f387845): GABA-A potentiation at gaba_scale 1.6-1.9 and excitability block at threshold_scale 1.24-1.36 and 1.44-1.56, with stage 4b's movement seeds and both of the parent's chip seed sets, the coarse levels rerun as a reproduction check. **Why:** stage 4b showed that in two families the headline rests on a single step of the dose grid, and a finer grid is the direct way to ask what happens inside it.

- **Outcome:** 3 of 4 (registration commit f387845). Excitability block resolves chip later against both chip runs; GABA-A potentiation stays chip later with the full index, and tied without the bitter-gated proboscis.
- **Revisit:** a fourth movement (for example a walking readout, once one exists) would tell whether GABA-A potentiation's lateness is specific to feeding.

D23. Stopping new science for the night (overnight)

- **Options:** further stages (each remaining branch needs a model not at hand or is expected to fail by construction); documents for the team.
- **Taken:** a slide outline for the deck (`report/SLIDES_OUTLINE.md`, every number cited), the README's document table brought up to date, and a morning summary in `PROGRESS.md`. **Why:** the headline has now been tested as far as the current bodies allow (stages 4a-4c), and the next scientific choices (D7 neuron volumes, a muscle-driven hind leg, a walking readout for drugs) are the team's to make.

D24. The volume size rule (user)

- **Options (D7's unexplored branch):** neuron volumes from a future release, or MANC volumes through the male-CNS-to-MANC matching; found instead: the male CNS v1.0 neuPrint neuron table (same bucket, same CC-BY release) has a voxel count per body.
- **Taken:** fetch that table whole (4.65 GB, source `malecns_sizes`), checked against the bucket's CRC32C, because the bucket publishes no MD5 for it **(user: full file, not a column-only range read)**; rerun stage 2h with the published rule (stage 2i, registration commit d03ee27) on fresh draws. A numpy CRC32C was written rather than adding a dependency.
- **Outcome:** 6 of 7 (registration commit d03ee27): the rhythm needs the circuit and promotor and remotor keep a reliable, non-antiphase offset; flexor-extensor alternation fails.
- **Revisit:** normalizing by the median over the whole nerve cord rather than over the network (stated choice: over the network, as for the synapse-count stand-in).

D25. Checking that stage 2i comes from the neurons' own volumes

- **Options:** report stage 2i as it stands; name the assumption most likely to make it wrong and test it (the project rule for results that look good).
- **Taken:** stage 2j (registration commit 1d1d644), fresh draws: the volume rule replicated, volumes shuffled across neurons (a fresh permutation per draw), and normalization by the nerve cord's median instead of the network's. **Why:** if shuffled volumes do as well, the result is about the spread of sizes, not the identified neurons; the normalization was a stated choice the published description does not fix.
- **Outcome:** 2 of 4 (registration commit 1d1d644). Shuffled volumes give no rhythm; the cord-median normalization changes nothing; the rhythm and the circuit's necessity replicate, but the promotor-remotor offset does not reach the registered reliability, so it is not claimed.
- **Revisit:** a registered test of the offset alone with more draws, if it matters for the paper.

D26. Walking as a drug readout (user)

- **Options (from D22's revisit):** a fourth movement for the drug design; the only walking readout at hand is stage 2i's fictive front-leg rhythm (rate model, volume sizes, no body).
- **Taken:** stage 4d (registration commit 73ffde6), fresh draws: the drug knobs applied to the walking network (weights scaled by presynaptic class, as in the spiking model; the excitability knob scales the activation threshold), each draw's DNg100 input found under vehicle and held fixed, onsets on stage 4c's combined dose series. **Why:** it asks whether GABA-A potentiation's lateness is specific to feeding and whether the headline holds for a rhythm.

- **Stated choice:** the threshold knob maps to the rate model's theta; fixed inputs, so a drug changes the response, not the stimulus.
- **Outcome:** 2 of 4 (registration commit 73ffde6). GABA-A and GluCl block break walking first; acetylcholine block is chip-first; GABA-A potentiation and the threshold knob leave walking intact; with walking as a fourth movement the chip is still late in every family.
- **Revisit:** a readout that respects the bimodal draws (the fraction of rhythmic draws) registered as its own test; a threshold mapping that also shifts the drive.

D27. Retesting the acetylcholine exception with a readout that respects the draws (user)

- **Options:** keep stage 4d's median readout; a fraction-of-rhythmic-draws readout; a per-draw paired analysis.
- **Taken:** the fraction readout on fresh draws (stage 4e, registration commit 180476f). **Why:** stage 4d's draws are bimodal, the median sits between the modes, and the exception entered the abstract; the fraction is the plainest readout of a bimodal outcome and uses the rhythm line registered since stage 2g.
- **Outcome:** 2 of 4 (registration commit 180476f). GABA-A and GluCl block break walking first; the acetylcholine exception does not replicate: mild block raises the share of rhythmic draws, and the two-sided index counts that as a change before the chip. The exception is withdrawn from the abstract, discussion, README and slides.
- **Revisit:** a one-sided walking index (only loss of rhythm counts) would be a new rule chosen after seeing this result; register it only with a reason that does not depend on the outcome.